# NITHUSHAN UTHAYARASA

Aspiring AI/ML Engineer
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LinkedIn | GitHub

## Professional Summary

Artificial Intelligence undergraduate at SLIIT with hands-on experience in machine learning, deep learning, computer vision, RAG, LLMs, and agentic AI. Skilled in developing intelligent systems, retrieval pipelines, and AI automation solutions and integrating AI into full-stack applications. Seeking an AI/ML engineering internship to apply technical and problem-solving skills to real-world projects.

## Education

**Sri Lanka Institute of Information Technology** Malabe
BSc (Hons) in Information Technology – Specialization in Artificial Intelligence 2024-2028
Third-year student | CGPA: 3.05 / 4.00

## Projects

### ContextIQ – RAG-Powered Intelligence System (RAG & LLMs)

- Built a multi-document RAG system with grounded answers, page-level citations, and conversational query handling.
- Implemented retrieval using Gemini embeddings, ChromaDB, BM25, RRF, TF-IDF reranking, and parent-child chunking.
- Developed a quantitative RAG evaluation pipeline using Hit@K, Recall@K, MRR, citation metrics, and ablation testing.
- Deployed the system with Streamlit, including retrieval inspection, context compression, performance tracking, and automated testing.
- Technologies: Python, RAG, LLMs, Vector Databases, Google Gemini, ChromaDB, Streamlit, PyMuPDF, BM25, TF-IDF
- GitHub: ContextIQ-RAG-System | Live Application: contextiq-ai-rag.streamlit.app

### Automated Manufacturing Inventory Coordinator (Agentic AI) – Team Project

- Built defect reporting and inventory roll quarantine and release features across ASP.NET Core, React, and Flutter.
- Developed a Validation and Safety agent to check suppliers, materials, PO calculations, and budgets before approval.
- Developed AI-assisted recommendations to identify inventory rolls and suggest quarantine using defect and stock data.
- Integrated validation into a four-agent LangGraph workflow, with results and audit history available on web and mobile applications.
- Technologies: Python, LangGraph, LangChain Core, FastAPI, Pydantic, C#, ASP.NET Core, React, Flutter, Dart, PostgreSQL
- GitHub: Automated Manufacturing Inventory Coordinator

### Garbage Image Classification (Deep Learning & Computer Vision)

- Developed a CNN transfer learning model to classify waste into 10 categories using over 20,000 images.
- Achieved 88.6% test accuracy through tuning, preprocessing, normalization, and train-validation-test evaluation.
- Applied preprocessing, augmentation, checkpointing, and prediction analysis to improve reliability and generalization.
- Evaluated the model using precision, recall, F1-score, a confusion matrix, and classification reports to identify errors.
- Technologies: Python, TensorFlow/Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook
- GitHub: SmartWaste Image Classification

### Cyber Threats & Financial Loss Prediction (Machine Learning)

- Built an end-to-end machine learning pipeline to predict high-risk financial losses from cyber incidents.
- Trained and compared Random Forest, CatBoost, XGBoost, LightGBM, and Extra Trees models with hyperparameter tuning.
- Performed feature engineering, preprocessing, cross-validation, and threshold optimization to improve prediction reliability.
- Evaluated models using accuracy, macro F1, recall, ROC-AUC, and feature importance; selected CatBoost as the best model.
- Technologies: Python, Pandas, NumPy, Scikit-learn, CatBoost, XGBoost, LightGBM
- GitHub: CyberThreat Financial Loss Prediction

## Technical Skills

- Programming – Python, Java, JavaScript
- AI / Machine Learning – Machine Learning, Deep Learning, NLP, Computer Vision, RAG, LLMs, Agentic AI, Vectors
- Frameworks & Libraries – TensorFlow/Keras, Scikit-learn, LangGraph, LangChain Core, OpenCV, Pandas, NumPy
- Backend & APIs – FastAPI, ASP.NET Core, Spring Boot, Node.js, Express.js, REST APIs, LLM APIs
- Databases & Retrieval – PostgreSQL, MongoDB, ChromaDB, Vector Databases, BM25, TF-IDF
- Software Engineering – OOP, Full-Stack Development, Database Design, Authentication & Authorization, JWT
- Tools – Git, GitHub, Streamlit, Postman, Jupyter Notebook, VS Code, IntelliJ IDEA, Vercel, Render
- Soft Skills – Problem Solving, Analytical Thinking, Continuous Learning, Communication, Adaptability

## Languages

English: Professional | Tamil: Native

## References

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